



**RV
UNIVERSITY**

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EDUCATIONAL INSTITUTIONS

SEMESTER – II

CS1006

Data Structures

Course Outline

2025-26

Course Faculty :

Prof. Kalpana Devi S

Dr. Dattatreya P Mankame

Prof. Vaidyanathan Sivasubramaniam

Dr. Nagaraja J

Prof. Narmadha T

Dr. Shouvik Chakraborty

Prof. Neha Najem

Prof. Vinod Kumar Raju

General Information

CS1006 – Data Structures	
Credits: 4, L T P: 2 - 1 - 2	
Internal Assessment- 70 marks	
Semester End Examination - 30 Marks	
Minimum Marks to pass	
Internal	Semester End
28	12

After completing the course, the students will be able to

- CO1:** Understand and explore the concepts of Linear and Non-Linear Data Structures to develop solutions for diverse problems.
- CO2:** Apply the concepts of Hashing, Sorting and Searching techniques to get an Optimized solution.
- CO3:** Evaluate and apply appropriate data structure(s) for real-world problems.
- CO4:** Demonstrate good coding practices involving lifelong learning.

Module – 1 Abstract Data Types & Memory Allocation

9 Hours

Strategies for choosing the appropriate data structure. Abstract data types. Stack Memory - Compile Time, Static memory allocation. Heap Memory - Runtime, dynamic memory allocation. Arrays – Revision of basic operations. Implementation of basic operations – Create, Insert, Traverse, Delete, Search and Update.

Module – 2 Linked List

9 Hours

Linked lists – Types – Singly Linked List – Doubly Linked List – Circular Linked List -(Circular Singly, Circular Doubly). Implementation of operations – Create, Insert, Traverse, Delete, Search and Update.

Module – 3 Stacks and Queues

9 Hours

Introduction to Stacks and Queues. Implementation of their basic operations. Use basic arrays & link list to create stack & queue data structures.

Module – 4 Sorting, Searching, and Hashing

9 Hours

Linear Search, Binary Search, Bubble Sort, Selection Sort, Insertion Sort. Hashing – Hash tables, has functions, strategies to avoid and resolve collisions.

Module – 5 Trees

9 Hours

Binary Tree, Binary Search Tree, AVL Tree, B-Trees, Common operations on these trees such as select min, select max, create, insert, delete. Perform various types of Tree Traversals.

Laboratory Component

To be completed before coming to lab:

- Create Hackerrank account (<https://www.hackerrank.com/>) using RVU email id
- Practice programs using any programming Platform
- Submission through Google classroom

Submission guidelines:

- i. Make sure to complete the submission on or before deadlines.
- ii. USN_Practical# (example: 1RVU23CSE002_Practical1.c)

Lab programs:

1. Program Execution – 6 Marks
2. Viva-Voce – 4 Marks

Total Marks – 10 Marks

List of Lab Programs

1. Programs to implement Recursion – Decimal to Binary , Printing Pyramid Pattern, Tower of Hanoi
2. Program to implement Arrays and its operations
3. Program to implement Stack and its Operations – Infix to Postfix Conversion, Postfix Evaluation
4. Program to implement Queue Data Structure – FCFS Scheduling Algorithm
5. Program to implement Linked List – Singly LL, Doubly LL, Singly Circular LL and Doubly Circular LL
6. Program to implement Binary Tree – Creation, Traversal , Insertion, Search, Deletion

7. Program to implement Search – Linear and Binary Search Algorithms
8. Program to implement Sorting –Insertion Sort
9. Program to implement Hash table using Linear Probing

Evaluation Scheme for Internal Assessment

**Failed in Internals (FI):
Overall Marks in Internal
Assessment < 28 Marks**

**Shortage of Attendance (SA):
Overall Attendance < 80%**

Note: The scheme is flexible and may change as per the university norms, if required.

S. No.	Component	Marks	Tentatively in
1	CP-I	20	02-03-2026 to 03-03-2026
	Objective Type Questions (Conceptual, Programming, etc.,)		
2	CP-II (Mid-Sem) Evaluation	25	30-03-2026 to 02-04-2026
	Pen- Paper		
3	CP-III	25	21-04-2026 to 23-04-2026
	Lab Record, Module Completion, Tutorials		

A student in FI/SA category needs to register for summer semester classes, attend the classes, and pass internal assessments.

Textbooks:

1. E. Horowitz and S. Sahni, *Fundamentals of Data Structures in C*, 2nd ed. Hyderabad, India: Universities Press, 2014, ISBN: 978-8173716058.
2. R. F. Gilberg and B. A. Forouzan, *Data Structures: A Pseudocode Approach with C*, 2nd ed. New Delhi, India: Cengage Learning India Pvt. Ltd., 2014, ISBN: 978-8131503140.
3. T. L. Kruse, *Data Structures and Program Design in C*, 2nd ed. New Delhi, India: Pearson India, 2001, ISBN: 978-8177584233.

Reference books:

1. J.P. Tremblay and P. G. Sorenson, *An Introduction to Data Structures with Applications*, 2nd ed. New Delhi, India: McGraw Hill, 2013.
2. A. M. Tenenbaum, *Data Structures Using C*, New Delhi, India: PHI Learning, 1989.

Thank You